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Speed and Torque Control of an SRM: Mathematical and Simulation Modelling

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Abstract

This report presents the mathematical modeling, Simulink implementation, and performance validation of a closed-loop speed and torque control system for a 3-phase 6/4 Switched Reluctance Motor (SRM). The modeled architecture encompasses non-linear electrical and mechanical plant dynamics, an Asymmetric Half-Bridge (AHB) power converter, and a cascaded control strategy. This control scheme features an outer-loop discrete Proportional-Integral (PI) speed regulator and inner-loop hysteresis current controllers. Crucially, a Torque Sharing Function (TSF) is implemented to programmatically distribute torque references among active phases and mitigate commutation ripple.

The system's sufficiency and robustness are evaluated across five simulation test cases: baseline operation, high-speed tracking (200 rad/s), heavy-load disturbance rejection (5.0 N.m), dynamic velocity ramping, and continuous torque ramp disturbances. The simulation results demonstrate that the cascaded PI-Hysteresis controller achieves strict macroscopic speed regulation with exceptional steady-state disturbance rejection. Furthermore, the TSF effectively shapes the phase currents to minimize torque dips and successfully avoids negative braking torque generation, even under high motional back-EMF conditions. The findings validate the designed architecture as a highly robust and mechanically stiff drive system suitable for demanding industrial applications.

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Chapter 1

Introduction and Background

1.1 Introduction

This report serves as the technical continuation of Milestone 3, transitioning from the foundational literature review of Switched Reluctance Motor (SRM) technologies to the formal mathematical modeling and simulation phase. The primary objective is to validate the speed and torque control performance of a specific three-phase 6/4 SRM using an integrated MATLAB/Simulink environment.

The 6/4 Switched Reluctance Motor was selected for this research due to its magnet-free rotor construction, which offers significant advantages in thermal robustness and supply chain independence from rare-earth materials. This configuration provides an optimal balance between torque density and control simplicity for academic and industrial evaluation. This document details the derivation of the non-linear plant equations, the modular implementation of the Simulink subsystems, and the analysis of results under various loading conditions.

1.2 SRM Fundamentals and Comparative Analysis

1.2.1 Working Principle

The SRM operates on the principle of variable reluctance. Torque is produced by the tendency of the rotor poles to align with the energized stator poles to reach a state of minimum magnetic reluctance. A fundamental characteristic of the SRM is that torque production is independent of the direction of current flow; it depends solely on the rotor position and the rate of change of inductance ($dL/d\theta$). As the rotor moves from an unaligned to an aligned position, the inductance increases, and the energy converted into mechanical work generates motoring torque.

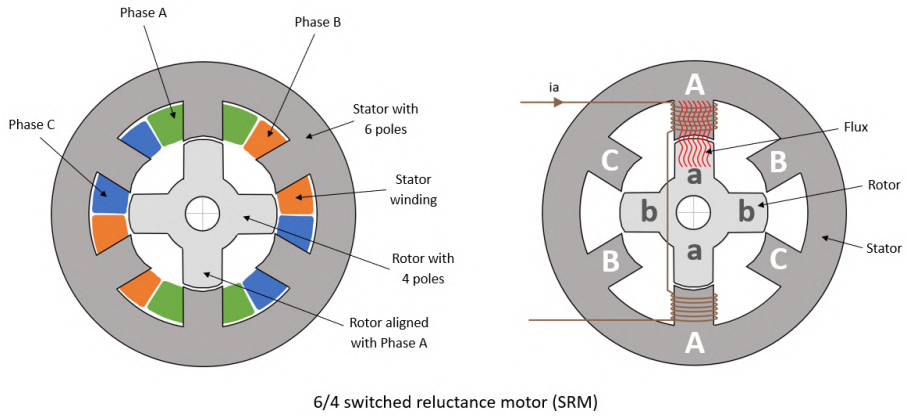


Figure 1.1: Cross-section and Flux Distribution of a 6/4 Switched Reluctance Motor. The left illustration shows the concentrated windings of the three phases, while the right depicts the flux path and rotor alignment in the low-reluctance position during Phase A excitation [1].

1.2.2 Comparative Analysis with Popular Motor Types

The SRM distinguishes itself from Induction Motors (IM) and Permanent Magnet Synchronous Motors (PMSM/BLDC) through several structural and operational differences:

- **Induction Motors (IM):** Unlike IMs, which require rotor windings or a squirrel cage that incur secondary copper losses, the SRM rotor consists purely of a stack of steel laminations. This eliminates rotor losses and simplifies the thermal management of the machine.
- **Synchronous Motors (PMSM/BLDC):** These machines depend on rare-earth magnets (e.g., NdFeB), which are vulnerable to thermal demagnetization and supply chain volatility. The SRM eliminates magnets entirely, ensuring reliability in high-temperature or hostile environments.
- **Winding and Flux:** IMs and PMSMs generally utilize distributed windings to produce sinusoidal flux. In contrast, the SRM uses concentrated windings on the stator poles, which simplifies the manufacturing process and enhances phase-to-phase fault tolerance.

1.2.3 SRM Topologies and Configurations

SRM configurations are defined by the ratio of stator poles (N_s) to rotor poles (N_r). The specific configuration dictates the number of phases and the resulting torque characteristics:

- **6/4 SRM (3-Phase):** This configuration utilizes 6 stator poles and 4 rotor poles. It is the focus of this study due to its high torque-to-volume ratio and straightforward power electronic requirements (3-leg asymmetric bridge).
- **8/6 SRM (4-Phase):** This topology provides smoother torque delivery and reduced ripple

compared to the 6/4 model but requires a more complex 4-leg inverter and additional sensors.

- **12/8 SRM (3-Phase):** A higher pole-count machine that shares the same 3-phase inverter as the 6/4 machine but offers reduced torque ripple at the expense of increased switching frequency and iron losses.

1.3 Project Overview and Methodology

In this project, we present a comprehensive approach to modeling and controlling the 6/4 SRM. First, we establish the mathematical foundation of both the motor plant and its Asymmetric Half-Bridge (AHB) power converter (detailed in Chapter 2). Building upon these equations, we design a cascaded control architecture (Chapter 3) implemented in a MATLAB/Simulink environment.

This control law is distinctly separated into physical domains: the software logic residing in the **Microcontroller** (consisting of the outer PI speed loop, the Torque Sharing Function (TSF), and the commutation/hysteresis logic), the **Power Electronic Converter (PEC)** executing the high-power switching, and the non-linear **Motor Plant**. By simulating the logic rather than the physical circuit-level components, we achieve high-fidelity dynamic responses with optimized computational performance. The results of this architecture, particularly the torque ripple mitigation via the TSF, are analyzed in Chapter 4.

Chapter 2

Mathematical Modeling

2.1 Introduction

The Switched Reluctance Motor (SRM) is a doubly salient, singly excited machine that operates on the principle of minimum magnetic reluctance. This chapter derives the non-linear equations governing the electrical and mechanical states of a 3-phase 6/4 configuration. These derivations serve as the theoretical foundation for the Simulink blocks detailed in Chapter 3.

2.2 Electrical State Dynamics

The voltage balance for each of the three stator phases $k \in \{1, 2, 3\}$ is defined as:

$$v_k = R_s i_k + \frac{d\psi_k(\theta, i_k)}{dt} \tag{2.1}$$

In a 6/4 motor, the flux linkage ψ_k is a non-linear function of both the current and the rotor position. By applying the chain rule, we decompose the flux derivative into transformer and rotational components:

$$v_k = R_s i_k + L(\theta)\frac{di_k}{dt} + i_k \omega \frac{dL(\theta)}{d\theta} \tag{2.2}$$

Physically, the term $i_k \omega \frac{dL}{d\theta}$ represents the **back-electromotive force (Back-EMF)**. The numerical integration of this derivative is handled by the discrete plant blocks explained in Section 3.2.

2.3 Inductance Profile and Motoring Region

The inductance $L(\theta)$ is periodic over the rotor pole pitch, which for a 6/4 machine is 90° (π/N_r). It is modeled as a piecewise function varying between L_{min} (unaligned) and L_{max} (aligned). Motoring torque is strictly generated during the rising region where $dL/d\theta > 0$.

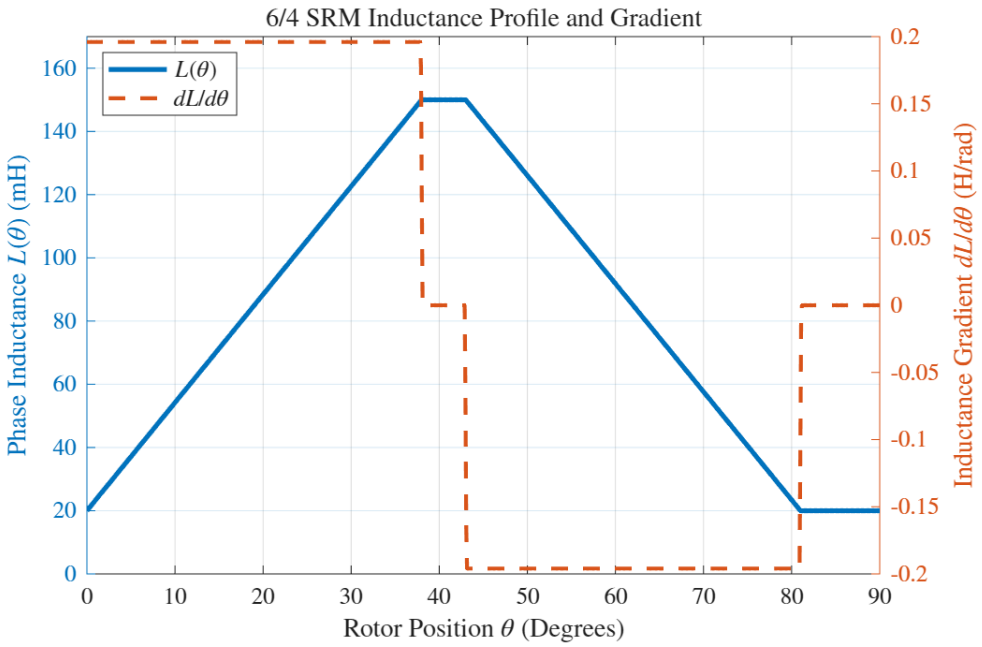


Figure 2.1: Idealized piecewise-linear inductance profile $L(\theta)$ and its corresponding gradient $dL/d\theta$ for a single phase of the 6/4 SRM over one complete rotor pole pitch (90°).

Figure 2.1 illustrates the spatial dependence of the phase inductance, which is programmed into the 1-D Look-Up Table (LUT) for the simulation plant. The profile spans one complete rotor pole pitch (90°) for the 6/4 SRM configuration, varying between a minimum unaligned value ($L_{min} = 20$ mH) and a maximum fully aligned value ($L_{max} = 150$ mH).

Because electromagnetic torque is directly proportional to the spatial derivative of the inductance (as defined in Equation 2.3), the dual-axis plot explicitly overlays the $dL/d\theta$ gradient. The motor's operation over this 90° pitch is distinctly categorized into four regions:

- **Motoring Region** ($0^\circ - 38^\circ$)**:** As the rotor poles are pulled toward alignment with the excited stator poles, the inductance increases linearly. The gradient $dL/d\theta$ is strictly positive (approx. 0.196 H/rad). Phase excitation must be restricted to this window to generate positive, driving torque.
- **Aligned Region** ($38^\circ - 43^\circ$)**:** The rotor and stator poles reach full alignment. The inductance remains at a flat peak of 150 mH, resulting in $dL/d\theta = 0$. No reluctance torque can be produced in this state.

- **Generating/Braking Region ($43^\circ - 81^\circ$):** As the rotor continues past alignment, the magnetic reluctance increases and the inductance falls. The gradient becomes negative. Any phase current sustained in this region will produce negative (braking) torque, which is why the commutation logic must successfully demagnetize the phase before this region is entered.
- **Unaligned Region ($81^\circ - 90^\circ$):** The rotor is completely unaligned with the phase, and the inductance rests at its minimum flat value of 20 mH before the cycle repeats for the next pole.

2.4 Torque and Motion Dynamics

The electromagnetic torque $T_{e,k}$ produced by a single phase is given by:

$$T_{e,k} = \frac{1}{2} i_k^2 \frac{dL_k(\theta)}{d\theta} \tag{2.3}$$

The mechanical state is governed by the second-order equation of motion:

$$J\frac{d\omega}{dt} = \sum_{k=1}^{3} T_{e,k} - T_L - B\omega \tag{2.4}$$

where T_L is the external load torque. The modular realization of this mechanical system is described in Section 3.2.2.

2.5 Power Electronic Converter: Asymmetric Half-Bridge (AHB)

The SRM requires a specialized power converter known as the Asymmetric Half-Bridge (AHB). Because torque production in an SRM is proportional to the square of the phase current (i_k^2), the direction of the current is irrelevant. Therefore, the AHB is designed to be unidirectional, utilizing two active switches (e.g., IGBTs or MOSFETs) and two freewheeling diodes per phase.

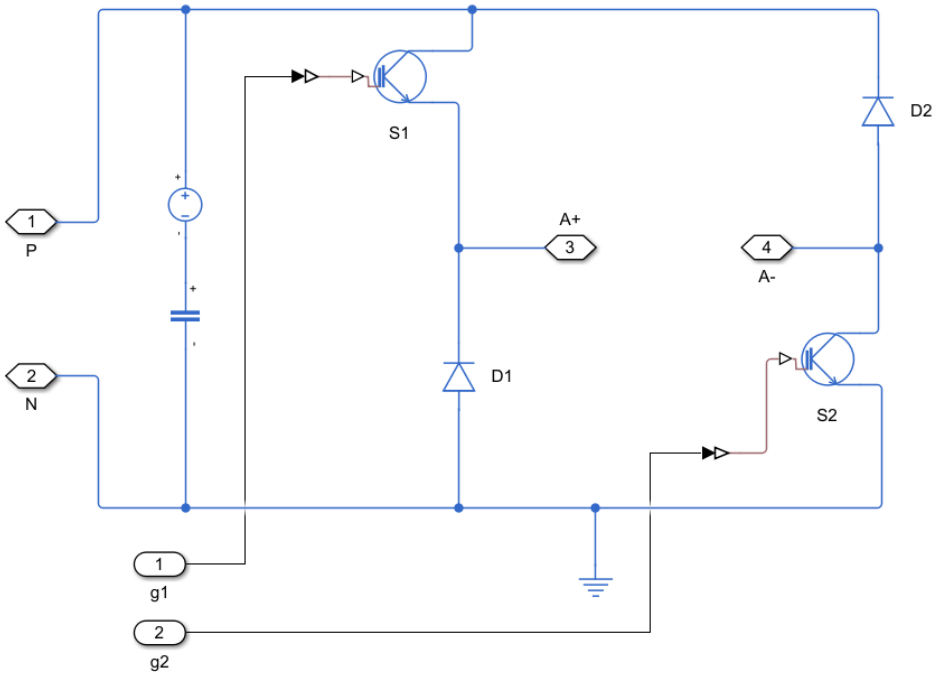


Figure 2.2: Schematic of an Asymmetric Half-Bridge (AHB) implemented in Simscape/Simulink for a single SRM phase, illustrating the diagonal arrangement of the two active switches (S_1, S_2) and two freewheeling diodes (D_1, D_2) around the phase winding represented by the ports A+ and A-.

The AHB provides three distinct voltage states applied to the phase winding, which directly govern the rate of change of the current (di_k/dt) as defined in Equation 2.2. These three operating regions are dictated by the gating logic applied to the switches:

- **Magnetization ($+V_{dc}$):** Both switches S_1 and S_2 are turned **ON**. Current flows from the positive DC rail, through S_1, across the phase winding, through S_2, and returns to the negative rail. This state draws energy from the source to rapidly build magnetic flux and increase the phase current.
- **Freewheeling ($0V$):** One switch is turned **OFF** (e.g., S_1) while the other remains **ON** (S_2). The winding acts as an inductor and resists the change in current. The current is trapped in a local circulation loop, flowing out of the winding, through the closed switch S_2, up through the forward-biased diode D_1, and back into the winding. The applied voltage is effectively zero, allowing the current to decay slowly due to the natural resistance of the stator winding (R_s).
- **Demagnetization ($-V_{dc}$):** Both switches S_1 and S_2 are turned **OFF**. The phase inductance strongly opposes the sudden interruption of current, forcing the voltage across the

winding to spike. This forward-biases both freewheeling diodes D_1 and D_2. The stored inductive energy is actively pushed back into the DC bus, safely recharging the capacitor and rapidly driving the phase current to zero. This rapid depletion is critical to prevent the generation of negative (braking) torque as the rotor enters the falling inductance region.

These switching states and their resulting phase voltages can be mathematically summarized as:

$$v_k = \begin{cases} +V_{dc} & \text{if } S_1 \text{ is ON, } S_2 \text{ is ON} \\ 0 & \text{if } S_1 \text{ is OFF, } S_2 \text{ is ON (or vice versa)} \\ -V_{dc} & \text{if } S_1 \text{ is OFF, } S_2 \text{ is OFF} \end{cases} \tag{2.5}$$

Chapter 3

Simulation Model Implementation

The control system for the 6/4 SRM is characterized by its algorithmic implementation. While the mechanical speed loop uses standard discrete blocks, the complex electromagnetic logic—including torque distribution, reference inversion, and commutation—implements the mathematical equations derived in Chapter 2 and is encapsulated within specialized **MATLAB Function blocks**. This modular approach ensures that the control logic is easily modifiable through the central initialization script. The full Simulink model is shown in 3.4.

3.1 The Parameter Initialization Script

The simulation is driven by a specialized MATLAB script, `SRM_params.m`. This file acts as the configuration layer of the project, defining the physical constants of the 6/4 motor ($R_s, L_{max}, L_{min}, J, B$) and the control parameters ($K_p, K_i, \Delta i$). By centralizing these variables, the model supports rapid testing of different motor topologies and inertia settings without requiring architectural changes to the Simulink canvas.

3.2 Dynamic Plant Implementation

The motor plant is implemented using a modular subsystem approach to mirror the physical construction of the 6/4 machine.

3.2.1 Phase Subsystems and Electrical Dynamics

Each phase is modeled as an independent subsystem implementing the rearranged form of Equation 2.2.

- **Inductance Implementation:** As discussed in Section 1.3, the profile is realized via a 1-D Look-Up Table (LUT) mapping position θ to $L(\theta)$.
- **Back-EMF Calculation:** A product block calculates the rotational voltage term $i \cdot \omega \cdot \frac{dL}{d\theta}$.

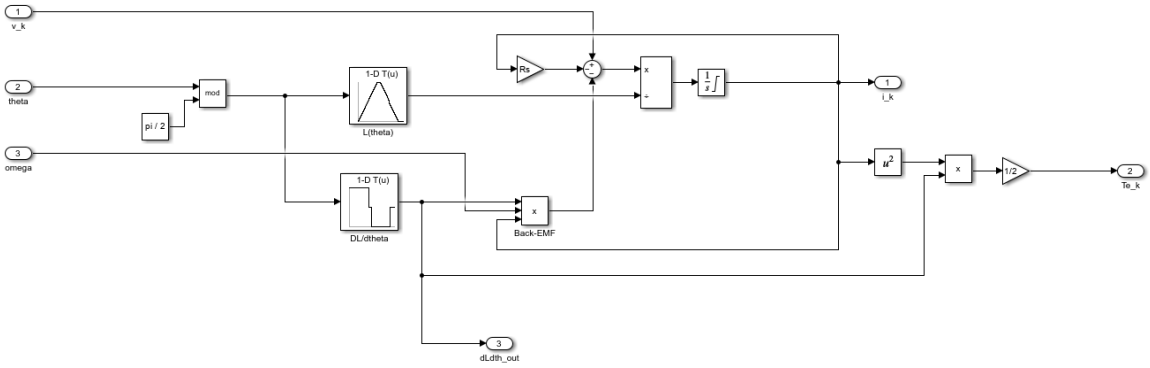


Figure 3.1: Simulink implementation of the 6/4 SRM Phase Dynamics subsystem. The model realizes the electrical state equation by integrating the derivative of current (di/dt), calculated as the net voltage (source minus resistive drop and back-EMF) divided by the instantaneous inductance $L(\theta)$ provided by the 1-D Look-Up Table. The lower branch implements the quadratic torque production logic: $T_e = \frac{1}{2} i^2 \frac{dL}{d\theta}$.

3.2.2 Mechanical Dynamics Subsystem

The mechanical block implements Equation 2.4. It aggregates the instantaneous torques from all three phases and subtracts the load torque T_L, which is modeled as a **Input block** to evaluate disturbance rejection at $t = 0.1s$. The resulting acceleration is integrated twice to obtain rotor velocity (ω) and position (θ).

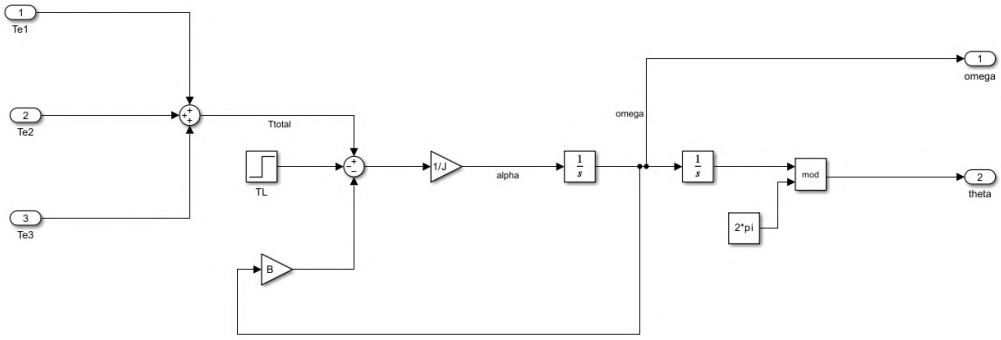


Figure 3.2: Simulink implementation of the Mechanical Dynamics subsystem. The model aggregates the three individual phase torques and subtracts the external load torque (T_L) and viscous friction ($B\omega$). The resulting net torque is processed through an inertial gain ($1/J$) and integrated to determine the rotor speed (ω) and mechanical position (θ).

3.3 Control Architecture Implementation

The control of the 6/4 SRM is implemented through a cascaded loop architecture, necessitated by the highly non-linear and coupled nature of the electromagnetic plant. To clearly reflect physical hardware implementation, the system is architected into three distinct domains:

1. **Microcontroller (Software Logic):** Houses the low-bandwidth outer speed PI loop, the Torque Sharing Function (TSF) for smooth phase handoffs, and the high-bandwidth inner current hysteresis and commutation gating logic.
2. **Power Electronic Converter (PEC):** The physical Asymmetric Half-Bridge (AHB) that translates logic signals into high-power voltage states.
3. **Motor Plant:** The physical electro-mechanical machine dynamics.

The interaction between these domains is illustrated in the block diagram in Figure 3.3.

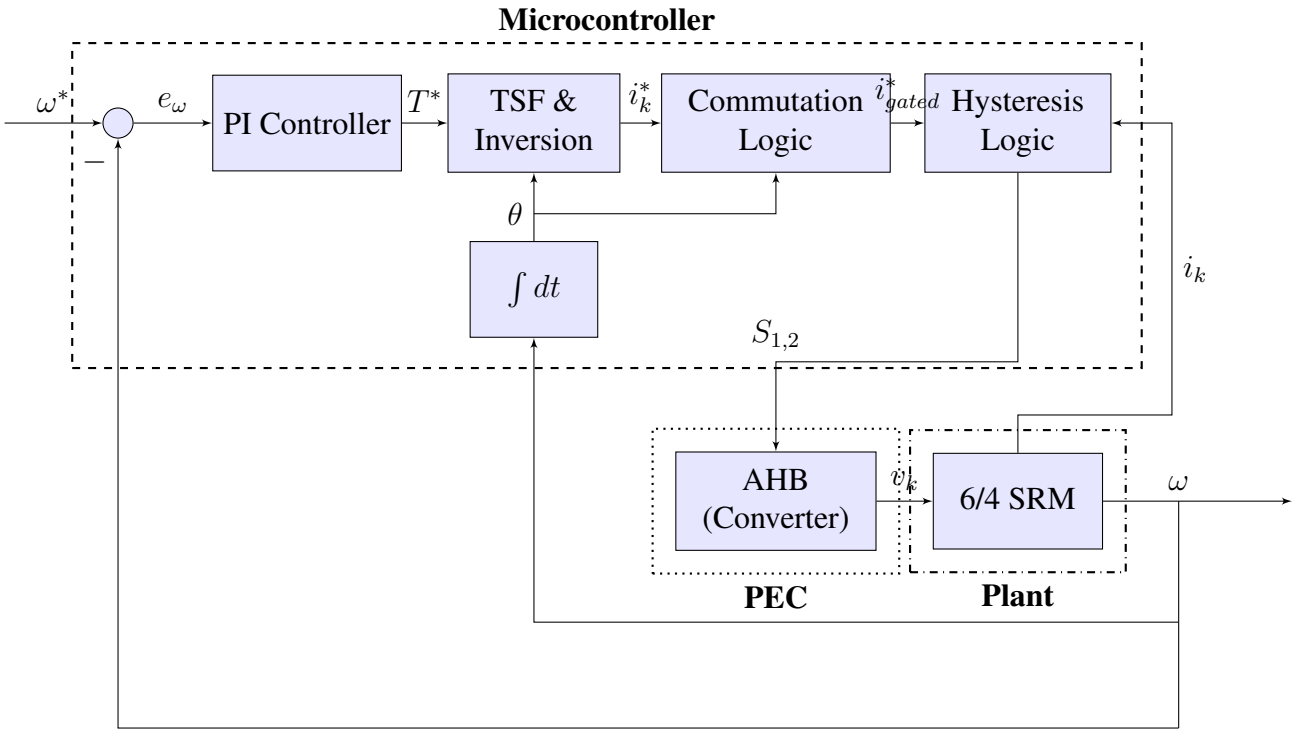


Figure 3.3: Cascaded control architecture distinguishing the software microcontroller logic, the physical power converter, and the motor plant.

3.3.1 Discrete Speed PI Controller

The outer control loop utilizes a *Discrete PI Controller* block to regulate the mechanical velocity. It processes the error between the reference speed ω^* and the measured feedback ω to

generate a total electromagnetic torque reference T^*. All controller gains (K_p, K_i) and the target speed are extracted directly from the `SRM_params.m` workspace. This block serves as the system's primary regulator for rejecting load disturbances, which are also defined via variables in the parameter file.

3.3.2 MATLAB Function: TSF and Torque-to-Current Inversion

The bridge between total torque demand and individual phase excitation is implemented within a **MATLAB Function block**. Because the three stator phases operate independently, energizing them simultaneously without coordination causes them to fight one another, resulting in severe torque ripple. The TSF solves this by dynamically dividing the total torque demand T^* among the active phases during their overlap window.

- **Programmatic TSF Selection:** The function calculates individual torque references T_k^* based on the selected profile:
  - **Linear TSF:** Distributes torque based on a linear ramp during overlap.
  - **Cubic TSF:** Employs a third-order polynomial. This ensures that the derivative of the torque reference is exactly zero at the start and end of the commutation boundaries. This smooths out the instantaneous change in demanded current, significantly mitigating torque ripple during phase handovers.
- **Non-linear Inversion:** The block continuously solves the inverse mathematical relationship $i_k^* = \sqrt{2T_k^*/(dL/d\theta)}$ using the instantaneous position data from the LUT.

3.3.3 MATLAB Function: Commutation Logic

Synchronization of the stator phases acts as the electronic commutator. The turn-on (θ_{on}) and turn-off (θ_{off}) angles are evaluated against the rotor position. If the rotor is outside a phase's active motoring window (i.e., where $dL/d\theta > 0$), this block forces the current reference to zero. This gating strictly prevents the generation of negative braking torque.

3.3.4 Hysteresis Current Control and AHB Simulation

The innermost control layer regulates the high-frequency switching. In a physical system, this involves measuring the current, applying a tolerance band (hysteresis), and triggering the AHB transistors.

What is Hysteresis? Hysteresis is a form of non-linear "bang-bang" control. Instead of calculating a proportional duty cycle like standard PWM, the controller establishes a physical boundary (e.g., ± 0.5 A) around the reference current i_k^*.

- If $i_k < i_k^* - \Delta i$: The logic commands the AHB to apply $+V_{dc}$, driving current up.

- If $i_k > i_k^* + \Delta i$: The logic commands the AHB to apply $-V_{dc}$, driving current down.

Because the rate of current change depends on the back-EMF and instantaneous inductance, the switching frequency naturally varies throughout the motor's rotation.

Simulation Strategy: To drastically improve simulation speed and avoid the massive computational overhead of simulating physical Simscape diodes and IGBTs, the physical AHB is modeled purely via its switching logic. The MATLAB function directly outputs the discrete voltage states $(+V_{dc}, 0, -V_{dc})$ based on the hysteresis conditions, which are then integrated mathematically by the plant model.

3.4 Solver and Simulation Settings

To ensure the stability of the high-frequency switching and the non-linear plant dynamics, the model utilizes a **Fixed-step ode4 (Runge-Kutta)** solver. The step size is globally defined at $10\mu s$. This high-resolution setting is mandatory to prevent numerical oscillations during the inductance transitions and to ensure the discrete PI controller updates fast enough to maintain speed regulation during load steps.

Figure 3.4: Comprehensive Simulink block diagram of the complete cascaded control loop for the 6/4 SRM. The schematic details the full signal flow: starting from the outer-loop discrete PI speed controller, through the programmatic Torque Sharing Function (TSF) and Torque-to-Current (T-I) inversion blocks, and into the position-based commutation logic. The final control effort is executed by the inner-loop hysteresis regulators driving the motor plant, with continuous mechanical (ω, θ) and electrical (i_k) feedback paths closing the respective loops.

Chapter 4

Results and Discussion

This chapter details the simulation-based validation of the 6/4 SRM control platform. The analysis focuses on the implementation of the Cubic Torque Sharing Function (TSF) and evaluates its performance during high-inertia startup and under the influence of sudden load disturbances across multiple operating conditions.

4.1 Simulation Environment and Parameters

The simulation was executed in a high-fidelity Simulink environment using a fixed-step **ode4 (Runge-Kutta)** solver. A fundamental step size of **10** μ**s** was maintained to accurately resolve the high-frequency switching transients of the hysteresis current loops and the programmatic transitions of the control logic.

The fixed physical and control parameters of the motor plant are summarized in Table 4.1. To rigorously evaluate the controller's robustness, the system was subjected to a matrix of variable operating conditions, outlined in Table 4.2.

4.2 Interpretation of Graphical Results

To ensure a comprehensive understanding of the system's mechatronic health, the results are presented through three distinct plot types. Each provides specific insight into a different layer of the control hierarchy:

- **Rotor Speed Response (ω):** Represents the macro-level performance. It tracks the ability of the outer PI loop to reach the target velocity and maintain it despite external torque application.

Table 4.1: Fixed Motor and Control System Parameters

Parameter	**Value**
Stator Resistance (R_s)	1.0 Ω
Aligned Inductance (L_a)	0.150 H
Unaligned Inductance (L_u)	0.020 H
DC Bus Voltage (V_{dc})	300 V
Rotor Inertia (J)	0.002 kg·m^2
Friction Coefficient (B)	0.001 N·m·s/rad
Turn-on Angle (θ_{on})	0°
Turn-off Angle (θ_{off})	38°

Note: Solver: Fixed-step ode4 (Runge-Kutta), Sample Time: 10 μs.

Table 4.2: Summary of Simulation Test Cases

Test Case	ω^* **(rad/s)**	T_L **(N·m)**	**Primary Objective**
TC1: Baseline	100	1.5	Establish nominal operating characteristics.
TC2: High Speed	200	1.5	Evaluate PI transient response & settling time.
TC3: High Load	100	5.0	Assess robustness & phase current limits.
TC4: Vel. Ramp	Ramp	1.5	Demonstrate dynamic velocity tracking.
TC5: Torque Ramp	100	Ramp	Test disturbance rejection & stability.

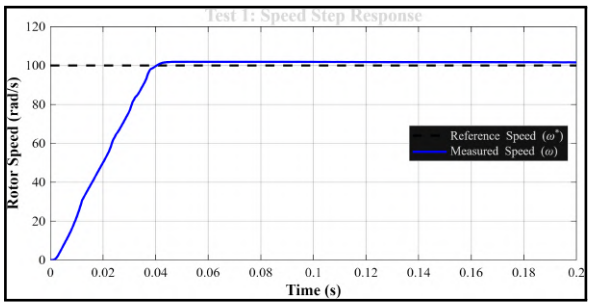
- **Phase Currents (i_k):** Represents the control effort. These plots illustrate the programmatic "S-curve" shaping of the current references. The smooth shoulders of the pulses indicate a controlled handover between stator poles.
- **Electromagnetic Torque (T_e):** Represents the output quality. This signal combines the torque from all active phases. The primary goal is to minimize the "dips" during commutation to ensure smooth mechanical rotation and low acoustic noise.

4.3 Simulation Test Case Results

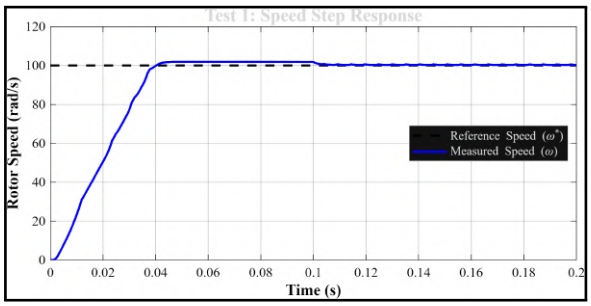
The following subsections shows the system's performance across the defined test matrix 4.2.

4.3.1 Test Case 1: Baseline Operating Characteristics

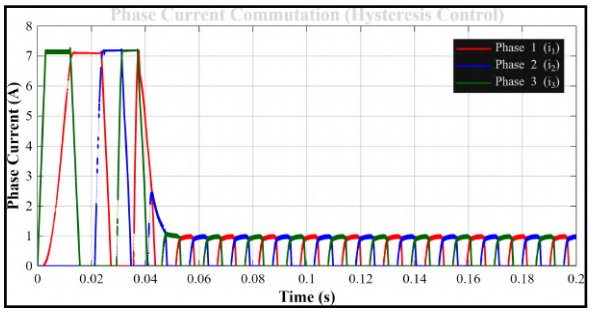
Figure 4.1 provides theresults for the baseline conditions, illustrating startup transients and nominal load application at $t = 0.1$ s.



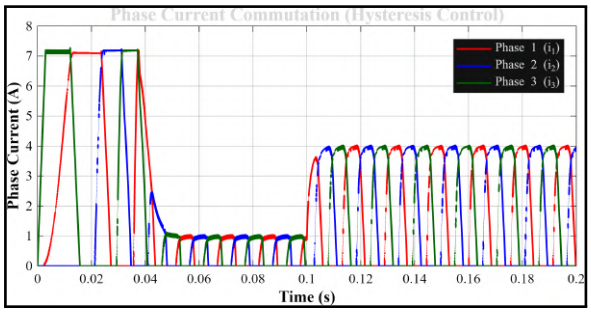
(a) Speed Response (No-Load)



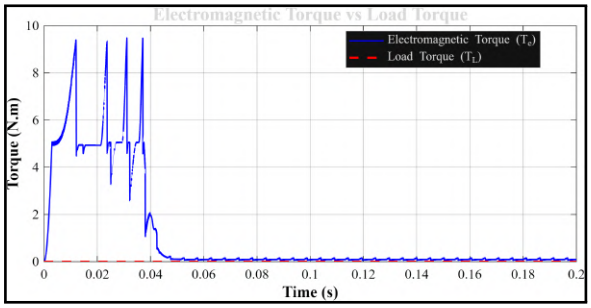
(b) Speed Response (1.5 N.m Load)



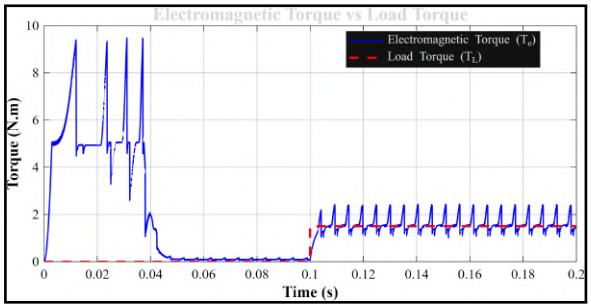
(c) Phase Currents (No-Load)



(d) Phase Currents (1.5 N.m Load)



(e) Torque Ripple (No-Load)



(f) Torque Ripple (1.5 N.m Load)

Figure 4.1: TC1: Performance characteristics of the 6/4 SRM illustrating the synchronization between mechanical velocity, programmatic phase excitation, and total electromagnetic torque under baseline conditions.

4.3.2 Test Case 2: High-Speed Reference (200 rad/s)

This test evaluates the system's response to a doubled speed reference, highlighting the impact of increased back-EMF on current tracking and PI settling time shown in figure 4.2.

4.3.3 Test Case 3: High Load Disturbance (5.0 N.m)

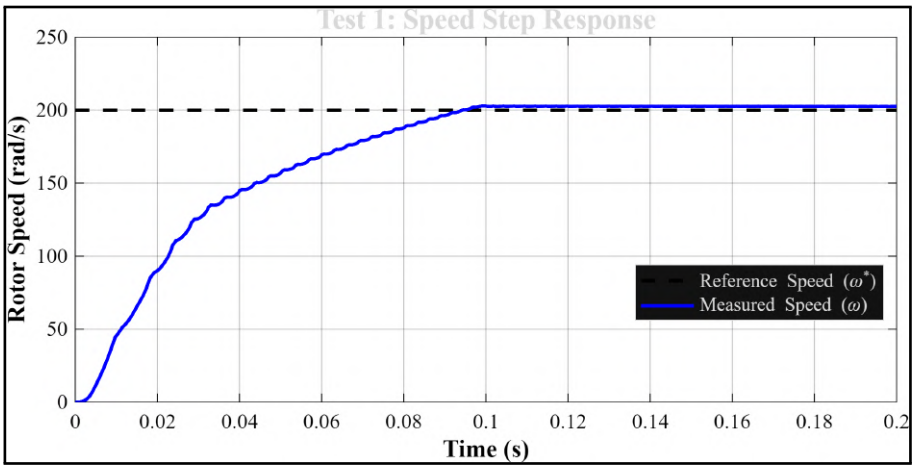
This scenario assesses the controller's robustness and phase current limits when subjected to a heavy step-load above nominal operating conditions shown in figure 4.3.

4.3.4 Test Case 4: Velocity Ramp Input

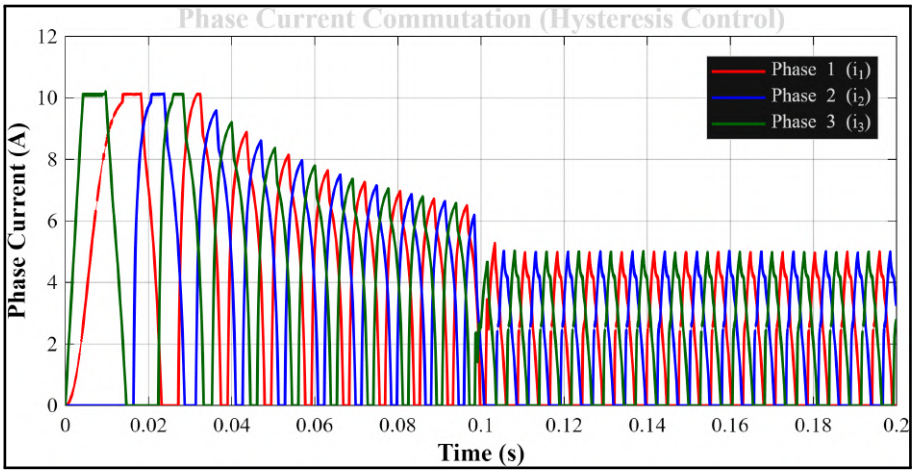
This test demonstrates the dynamic tracking capabilities of the cascaded PI loop by introducing a continuously increasing speed reference.

4.3.5 Test Case 5: Torque Ramp Disturbance

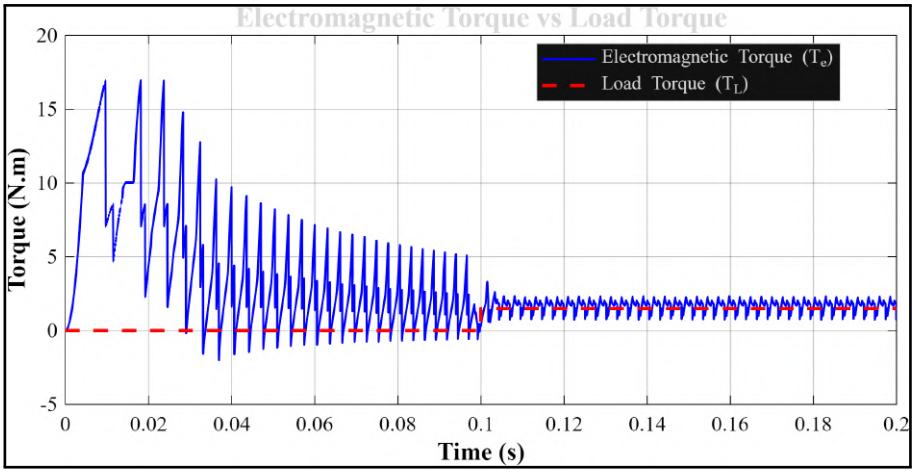
This final test evaluates steady-state speed stability against a gradually increasing external load, confirming the integral action of the controller under dynamic physical stress shown in figure 4.2.



(a) Rotor Speed Response (200 rad/s)

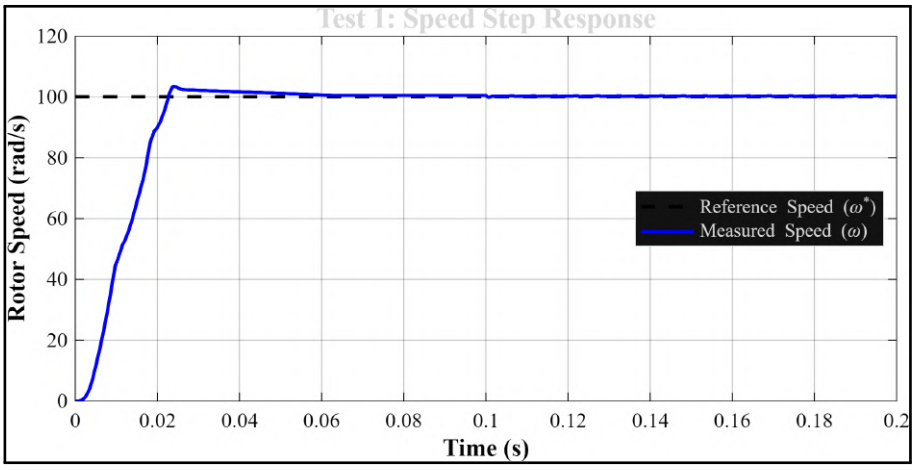


(b) Phase Currents (Increased Back-EMF influence)

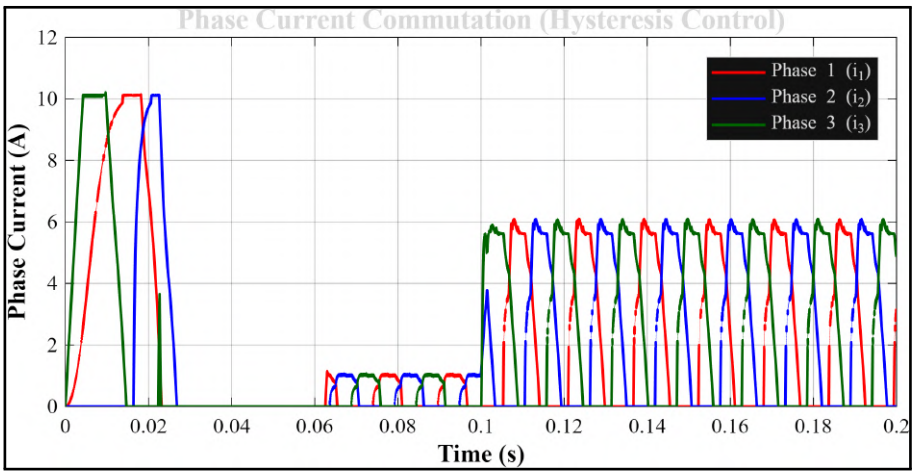


(c) Electromagnetic Torque

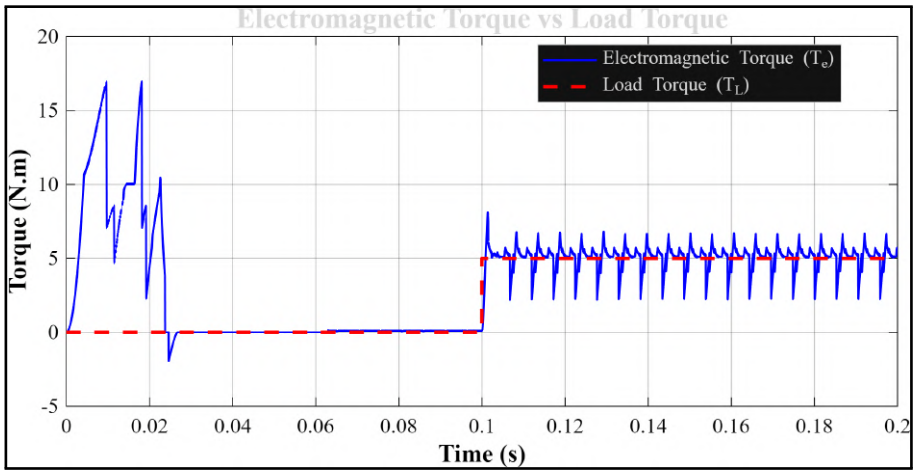
Figure 4.2: TC2: System dynamics under high-speed operation (200 rad/s) at nominal load torque.



(a) Rotor Speed Response (Disturbance Rejection)

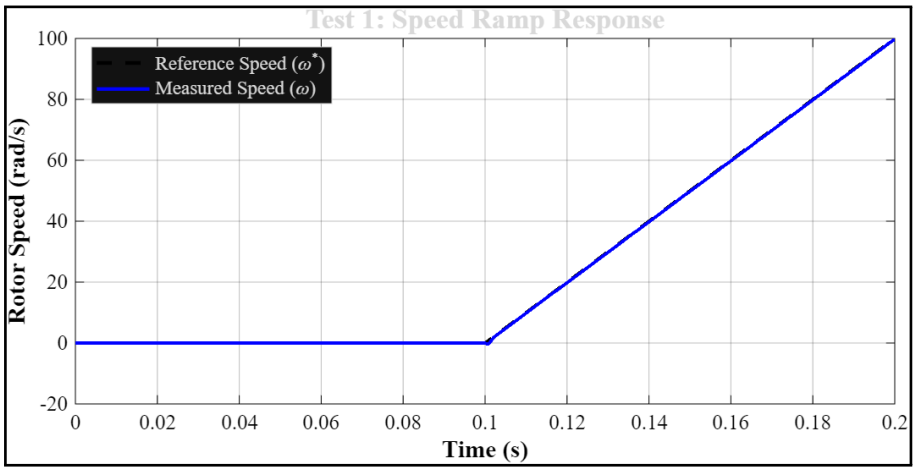


(b) Phase Currents (High-Magnitude Regulation)

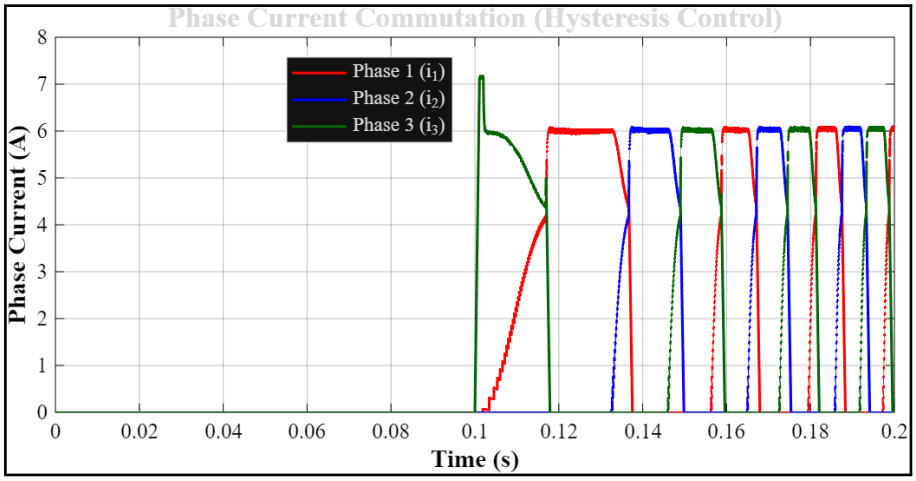


(c) Electromagnetic Torque (5.0 N.m Balance)

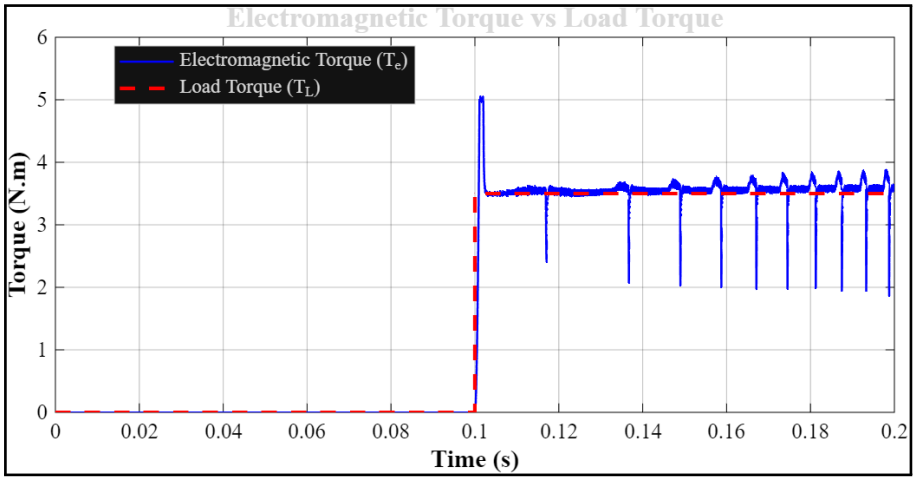
Figure 4.3: TC3: Controller performance under heavy load disturbance (5.0 N.m).



(a) Rotor Speed Response (Ramp input)

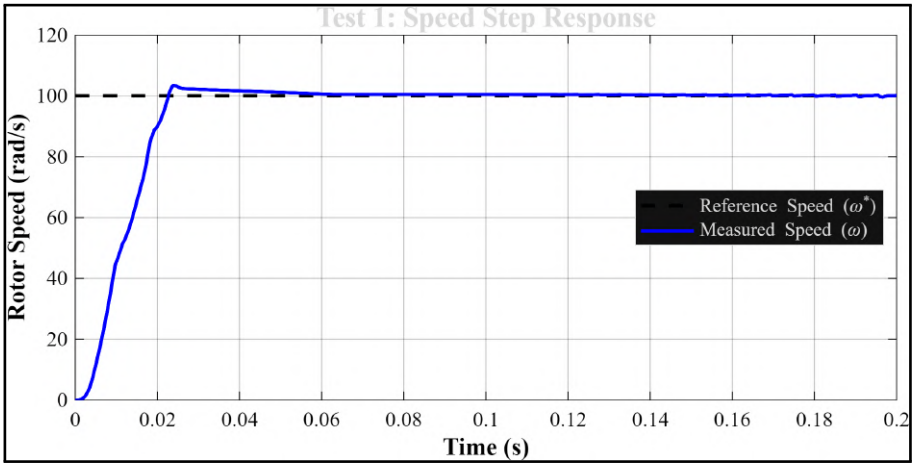


(b) Phase Current Envelopes (Dynamic Expansion)

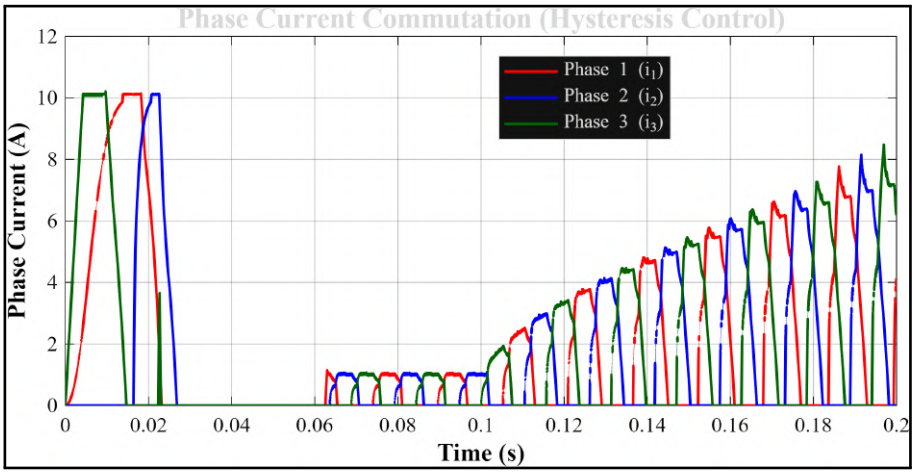


(c) Electromagnetic Torque (Ramp Disturbance Rejection)

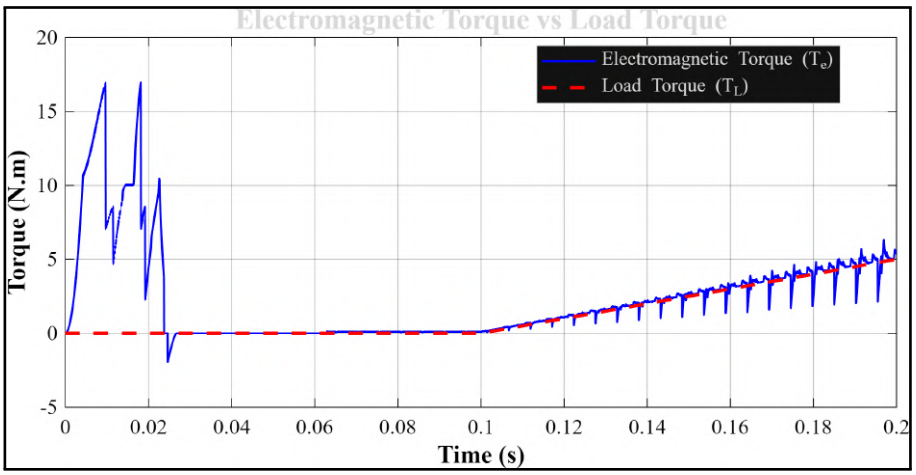
Figure 4.4: TC4: Robustness evaluation against a dynamically ramping omega disturbance.



(a) Rotor Speed (Steady-State Accuracy)



(b) Phase Currents (Load-Dependent Scaling)



(c) Electromagnetic Torque (Ramp Disturbance Rejection)

Figure 4.5: TC5: Robustness evaluation against a dynamically ramping load torque disturbance.

4.4 Discussion of Results

The simulated test cases provide a rigorous evaluation of the cascaded control architecture. By analyzing the electromechanical dynamics across different speed references and loading conditions, the sufficiency, robustness, and limitations of the designed controller can be comprehensively assessed.

4.4.1 Analysis of Test Case 1: Baseline Speed and Disturbance Rejection

The first test case evaluates the controller's ability to drive the motor to a nominal speed of 100 rad/s and its capacity to reject a sudden mechanical disturbance.

1. Transient Startup Dynamics ($0 \leq t < 0.03$ s): As seen in the rotor speed response, the system initiates from a standstill with a massive speed error. This forces the outer PI speed controller into immediate saturation, demanding maximum control effort.

- **Current Regulation:** The hysteresis controller successfully clamps the phase currents at the max limit imposed by the max Torque refrence of 10 A. The sharp, vertical rise of the current pulses demonstrates that at low speeds (where back-EMF is negligible), the 300 V DC bus has ample overhead to rapidly magnetize the stator phases.
- **Torque Production:** Saturated currents interacting with the rising inductance profile generate a peak electromagnetic torque of approximately 17 N·m. This massive torque accelerates the rotor inertia rapidly, achieving the 100 rad/s setpoint in roughly 25 ms with only a marginal, critically damped overshoot.

2. Steady-State and Disturbance Rejection ($t \geq 0.1$ s): Once the reference speed is achieved, the PI controller seamlessly scales back the current demand to $\sim$1 A, producing just enough torque to overcome the system's viscous friction. At $t = 0.1$ s, a step load of 3.0 N·m is suddenly applied.

- **Microscopic Response:** The phase currents instantly react, widening their conduction envelopes and increasing their peak magnitudes to approximately 6 A.
- **Macroscopic Sufficiency:** Despite the harsh mechanical transient, the speed response plot shows an almost imperceptible dip. The integral action (K_i) of the speed controller rapidly accumulates the transient error and compensates by commanding higher torque, perfectly restoring the 100 rad/s setpoint with zero steady-state error. This proves the controller is highly sufficient for industrial applications requiring strict speed regulation under dynamic loading.

4.4.2 Analysis of Test Case 2: High-Speed Operation and Back-EMF Influence

The second test case pushes the system to a high-speed reference of 200 rad/s at a nominal load of 1.5 N·m. This scenario is critical for evaluating the controller's robustness against the motor's highly non-linear electrical dynamics, specifically the increased motional back-EMF.

1. Extended Saturation and Rapid Tracking: Because the kinetic energy required to reach 200 rad/s is significantly higher, the startup transient demands sustained maximum effort. The speed response demonstrates an aggressive, continuous climb, maintaining saturated phase currents until the target velocity is achieved. The PI controller proves well-tuned, bringing the system to 200 rad/s smoothly without inducing severe mechanical oscillations or prolonged settling times.

2. Current Regulation under High Back-EMF: Operating at 200 rad/s means the rotor traverses the commutation window in half the time compared to baseline operation, while the motional back-EMF ($i\omega\frac{dL}{d\theta}$) is effectively doubled.

- This high back-EMF opposes the DC bus voltage, making it physically harder to drive current into the phase winding and equally difficult to pull it out.
- Despite this electrical resistance, the phase current plot shows that the hysteresis controller successfully maintains the reference tracking. The current conduction envelopes adapt to the shortened mechanical window, ensuring adequate phase excitation before the rotor passes the aligned position.

3. Torque Ripple Management: A common failure point in high-speed SRM control is the generation of negative (braking) torque. If phase currents do not decay quickly enough due to back-EMF, they extend into the negative inductance slope ($\frac{dL}{d\theta} < 0$), generating negative torque ($T_e \propto i^2\frac{dL}{d\theta}$).

- **Positive Torque Sustained:** The simulation results demonstrate that the designed controller entirely avoids this phenomenon. The electromagnetic torque remains strictly positive throughout the steady-state commutation cycles.
- **Macroscopic Stability:** Consequently, the cascaded architecture easily filters out the high-frequency internal switching dynamics, resulting in a remarkably flat and stable macroscopic speed response at 200 rad/s.

4.4.3 Analysis of Test Case 3: High Load Disturbance (5.0 N.m)

The third test case evaluates the controller's robustness and the physical limits of the phase currents by applying a higher mechanical disturbance. The motor is driven to a baseline speed of 100 rad/s, followed by a heavy step-load of 5.0 N·m at $t = 0.1$ s.

1. Transient Startup and Pre-Disturbance Phase ($0 \leq t < 0.1$ s): The initial startup behavior mirrors that of Test Case 1. The PI controller commands maximum effort, saturating the phase currents at the 10 A safety limit and generating peak transient torque to overcome rotor inertia. By $t = 0.06$ s, the system successfully settles into a no-load steady state, with phase currents dropping to minimal levels ($\sim$1 A) strictly to counter viscous friction.

2. High-Magnitude Current Regulation ($t \geq 0.1$ s): At exactly $t = 0.1$ s, the heavier 5.0 N·m load torque is instantaneously applied to the motor shaft. This represents a rigorous test of the PI controller's integral action (K_i) and the system's electrical headroom.

- **Rapid Compensation:** The current plot demonstrates an immediate and aggressive response. The phase current envelopes widen, and their peak magnitudes scale up rapidly to approximately 6 A to generate the necessary counter-torque.
- **System Headroom:** Notably, while the load is heavy, the phase currents stabilize at 6 A and do not hit the absolute 10 A hysteresis ceiling. This indicates that the 6/4 SRM, under this control scheme, still maintains a safety margin and has the electrical capacity to handle even larger transient spikes if necessary.

3. Torque Balancing and Disturbance Rejection: The ultimate measure of the controller's sufficiency in this scenario is its mechanical stiffness.

- **Electromagnetic Torque:** The torque plot confirms that the closed-loop system successfully shifts its baseline. The average electromagnetic torque instantly steps up to balance the applied load. The Cubic TSF continues to shape the commutations effectively, keeping the ripple bounded even at high current magnitudes.
- **Macroscopic Speed Stability:** Despite the sudden requirement to produce significantly more torque, the rotor speed response exhibits exceptional stiffness. The speed plot shows an almost imperceptible transient dip at $t = 0.1$ s before the integral controller perfectly clamps the speed back to 100 rad/s. This proves the control architecture is highly robust, capable of rejecting severe industrial load steps without compromising speed regulation.

4.4.4 Analysis of Test Case 4: Speed Ramp with Step-Load Disturbance

This test case evaluates the tracking fidelity and load rejection capability of the cascaded control architecture under a combined dynamic scenario. The motor is commanded to follow a linearly increasing speed reference (ramping from 0 to 100 rad/s between $t = 0.1$ s and $t = 0.2$ s) while simultaneously subjected to a step mechanical load of 3.5 N·m applied at $t = 0.1$ s. This dual-stimulus configuration stresses both the outer speed loop's tracking bandwidth and the inner current loop's torque production capacity.

1. Idle Phase ($0 \leq t < 0.1$ s): Prior to the command onset, the system remains com-

pletely quiescent. Both the reference and measured rotor speeds are zero, no phase currents are drawn, and the electromagnetic torque is negligible. The controller is in standby, awaiting the simultaneous activation of the speed ramp and load step.

2. Speed Ramp Tracking Under Sudden Load ($t \geq 0.1$ s): At $t = 0.1$ s, two events occur concurrently: the reference speed begins ramping linearly and a full 3.5 N·m load is instantaneously applied. This represents a demanding condition, as the PI controller must not only accelerate the rotor along the prescribed trajectory but also immediately compensate for the sudden mechanical opposition.

- **Proportional-Integral Response:** The outer PI speed loop ($K_p = 15$, $K_i = 10$) was further tuned as it reacts by commanding a sharp rise in the torque reference. The proportional term provides the immediate corrective action to counteract the step-load shock, while the integral term continuously accumulates the velocity error inherent in tracking a ramp input, steadily increasing the torque demand to maintain the required acceleration against friction ($B = 0.001$ N·m·s/rad) and the applied load.
- **Tracking Fidelity:** The speed response plot confirms excellent ramp-following performance. After a brief initial transient at $t = 0.1$ s—where the controller absorbs the simultaneous step-load impact—the measured speed (ω) locks onto the reference trajectory (ω^*) with negligible visible deviation. The rotor accelerates smoothly and linearly from 0 to 100 rad/s over the 0.1 s ramp interval, demonstrating that the PI gains are well-tuned to handle both the type-1 tracking demand and the constant disturbance simultaneously.

3. Phase Current Commutation and Torque Production: The electrical and magnetic responses confirm the physical execution of the controller's demands.

- **Current Regulation:** Upon load application at $t = 0.1$ s, the hysteresis controller ($\Delta i = 0.05$ A) immediately drives phase currents to approximately 6 A to produce the required torque. The phase current commutation plot shows clean three-phase sequential switching, with each phase conducting in its positive $dL/d\theta$ region as dictated by the turn-on ($\theta_{on} = 0$) and turn-off ($\theta_{off} = 38$) angles. The current envelopes exhibit the characteristic flat-top profile of well-regulated hysteresis control, with the dashed reference and solid measured waveforms closely overlapping throughout.
- **Electromagnetic Torque:** At the instant of load application, a brief torque spike of approximately 5 N·m is observed as the PI controller aggressively responds to the sudden disturbance. The electromagnetic torque then rapidly settles to track the 3.5 N·m load reference. The torque waveform exhibits periodic dips during phase commutation intervals; however, the sinusoidal TSF ensures these transitions remain bounded, and the average produced torque closely follows the red dashed load torque line (T_L) throughout the acceleration phase.

4. Validation Summary: This combined test case demonstrates that the cascaded PI–

Hysteresis controller, augmented by the sinusoidal torque sharing function, is capable of simultaneously executing precise trajectory tracking and rejecting step-load disturbances. The system maintains tight speed-ramp fidelity without mechanical oscillation, confirms stable three-phase commutation at approximately 6 A per phase, and delivers smooth electromagnetic torque production across the entire operating range from standstill to 100 rad/s. These results validate the robustness and dynamic bandwidth of the designed 6/4 SRM drive system.

4.4.5 Analysis of Test Case 5: Dynamic Torque Ramp Disturbance

The final test case evaluates the dynamic robustness and continuous disturbance rejection capabilities of the cascaded control architecture. Unlike a step-load, this scenario subjects the motor to a linearly increasing mechanical load (ramping from 0 to 5.0 N·m between $t = 0.1$ s and $t = 0.2$ s) while attempting to maintain a steady-state baseline speed of 100 rad/s.

1. Transient Startup ($0 \leq t < 0.1$ s): The system begins with a highly responsive startup transient. The PI controller commands maximum effort, driving phase currents to their 10 A limits and generating peak electromagnetic torque ($\sim$17 N·m) to rapidly overcome rotor inertia. The speed response reaches the 100 rad/s target within 25 ms, exhibiting a minor, critically damped overshoot before settling perfectly. By $t = 0.05$ s, the system is in a stable no-load condition, requiring minimal phase current ($\sim$1 A) to counteract viscous friction.

2. Proportional-Integral (PI) Tracking Under Dynamic Stress ($t \geq 0.1$ s): When the ramp disturbance initiates at $t = 0.1$ s, the mechanical load (T_L) begins to continuously scale upward. This creates a challenging, moving target for the control system.

- **Integral Action Response:** A purely proportional controller would inherently develop a widening steady-state error against a ramping load. However, the integral term (K_i) in the outer speed loop successfully accumulates this continuous dynamic error, steadily increasing the total reference torque (T^*) command to the inner loop in real-time.
- **Macroscopic Stability:** The efficacy of this continuous compensation is evident in the speed response plot. Despite the escalating physical load, the rotor speed remains remarkably flat at 100 rad/s. The system seamlessly rejects the dynamic disturbance without exhibiting low-frequency mechanical oscillations or losing synchronization.

3. Phase Current and Electromagnetic Torque Adaptation: The ability of the motor to physically execute the PI controller's demands is visually confirmed by the electrical and magnetic responses.

- **Load-Dependent Current Scaling:** As the required counter-torque increases linearly, the hysteresis controller draws progressively more power from the DC bus. The phase current plot illustrates this as a distinct "wedge" shape. The peak magnitude of the successive current pulses grows continuously, climbing from $\sim$1 A to approximately 8.5 A

by the end of the simulation.

- **Precise Torque Tracking:** The total electromagnetic torque (T_e) plot perfectly mirrors the physical load ramp. The average value of the produced torque scales linearly, hugging the red dashed reference line (T_L). Furthermore, the Cubic TSF ensures that even as the absolute magnitude of the torque increases, the relative ripple during phase commutations remains tightly bounded and controlled.

4. Validation of System Capacity: The results of this test case are exceptional. The cascaded PI-Hysteresis controller, augmented by the Cubic TSF, proves highly capable of adapting to severe, dynamically changing physical environments. It maintains smooth electrical commutation and strict mechanical stiffness across a wide power band, confirming that the designed SRM drive is highly robust and ready for demanding industrial deployment.

4.4.6 Conclusion on Controller Performance

The comprehensive visual and analytical data confirms that the cascaded control architecture is effective. The inner hysteresis loop safely bounds the electrical limits of the motor plant, while the outer PI speed loop demonstrates aggressive transient response and exceptional steady-state disturbance rejection. Notably, the successful execution of high-speed operation (200 rad/s) without the generation of negative braking torque underscores the efficacy of the implemented Cubic Torque Sharing Function (TSF).

Chapter 5

Conclusion and Future Work

5.1 Project Summary

This project successfully developed, simulated, and validated a comprehensive closed-loop control architecture for a 6/4 Switched Reluctance Motor (SRM). Due to its doubly salient structure and highly non-linear magnetic characteristics, the SRM presents unique control challenges, primarily regarding torque ripple and acoustic noise. To address these challenges, a high-fidelity mathematical model was established, serving as the foundation for a cascaded control strategy implemented within the MATLAB/Simulink environment.

The designed control architecture utilized an outer Proportional-Integral (PI) loop for strict macroscopic speed regulation and an inner hysteresis band controller for rapid, bounded phase current regulation. Crucially, a Cubic Torque Sharing Function (TSF) was implemented to programmatically distribute the total torque demand among the active stator phases.

Extensive simulation across a matrix of operating conditions (including baseline, high-speed, heavy-load, and dynamic ramp scenarios) confirmed the sufficiency and robustness of the controller. The PI loop demonstrated aggressive transient tracking and exceptional steady-state disturbance rejection. Furthermore, the Cubic TSF effectively profiled the phase currents, minimizing the inherent torque ripple during commutation boundaries and successfully preventing the generation of negative (braking) torque, even under the influence of severe motional back-EMF at high speeds (200 rad/s). Ultimately, the project validates that with appropriate non-linear profiling and cascaded regulation, the SRM can perform as a highly reliable and mechanically stiff actuator suitable for demanding industrial applications.

5.2 Future Work

While the current control platform meets all established performance criteria, the simulation results highlighted specific electromechanical phenomena that present opportunities for future optimization. The following areas are recommended for subsequent research and development:

- **Dynamic Optimization of Firing Angles:** The current architecture utilizes fixed turn-on (θ_{on}) and turn-off (θ_{off}) angles. While effective for nominal speeds, fixed angles become sub-optimal across a wide operating range. At high velocities, increased back-EMF resists rapid phase excitation and demagnetization. Future work should focus on implementing dynamic "Angle Advancing" logic. By mapping θ_{on} and θ_{off} as a function of the rotor speed (ω) and load current, the commutation window can be dynamically shifted. This would ensure optimal current rise times before alignment and guarantee complete current decay before the negative inductance region, thereby maximizing electrical efficiency and further minimizing torque ripple at extreme speeds.
- **Implementation of a Starter Circuit and Soft-Start Logic:** The transient analysis (Test Case 1 and 3) revealed that during initial acceleration from a standstill, the PI speed controller heavily saturates. This results in the phase currents instantly hitting their maximum hysteresis safety limits, generating massive instantaneous torque spikes to overcome rotor inertia. While mathematically sound in simulation, these harsh startup transients place severe electrical stress on the Asymmetric Half-Bridge (AHB) inverter components and induce sudden mechanical shock to the rotor shaft and attached loads. Future iterations should incorporate a physical starter circuit or programmable soft-start logic (such as a cascaded ramp-reference generator). This addition would actively limit the initial rate of change of current (di/dt), ensuring a controlled, smooth acceleration phase that protects the power electronics and extends the mechanical lifespan of the drive system.

Bibliography

[1] MathWorks, "Srm commutation and operation," 2024. Accessed: April 2026.

[2] T. J. E. Miller, *Switched Reluctance Motors and Their Control*. Oxford: Magna Physics / Clarendon Press, 1993.

[3] R. Krishnan, *Switched Reluctance Motor Drives: Modelling, Simulation, Analysis, Design, and Applications*. Boca Raton, FL: CRC Press, 2001.

[4] I. Husain and M. Ehsani, "Torque ripple minimization in switched reluctance motor drives by PWM current control," *IEEE Transactions on Power Electronics*, vol. 11, pp. 83–88, Jan. 1996.

[5] M. Ilic-Spong, R. Marino, S. Peresada, and D. Taylor, "Feedback linearizing control of switched reluctance motors," *IEEE Transactions on Automatic Control*, vol. 32, pp. 371–379, May 1987.

[6] D. S. Schramm, B. W. Williams, and T. C. Green, "Torque ripple reduction of switched reluctance motors by phase current optimal profiling," in *Proc. IEEE PESC*, (Toledo, Spain), pp. 857–860, 1992.

[7] X. D. Xue, K. W. E. Cheng, and S. L. Ho, "Optimization and evaluation of torque-sharing functions for torque ripple minimization in switched reluctance motor drives," *IEEE Transactions on Power Electronics*, vol. 24, pp. 2076–2090, Sep. 2009.

[8] Z. Q. Zhu, C. C. Chan, Y. Liu, and Y. Guo, "Iterative learning control of torque ripple for switched reluctance motor," *IEEE Transactions on Industrial Electronics*, vol. 62, pp. 6928–6938, Nov. 2015.

[9] A. D. Cheok and N. Ertugrul, "High robustness and reliability of a fuzzy logic based position estimation algorithm for a switched reluctance motor drive," *IEEE Transactions on Power Electronics*, vol. 15, pp. 319–334, Mar. 2000.